

Advanced Assignment: Build a ReAct Travel Assistant Agent

Course Context

You have already learned:

- AI Agents
- Agentic AI
- LangChain fundamentals
- Custom tools
- ReAct agents
- LangGraph basics
- Agent Executor workflow

This assignment is **industry-level** and requires students to build a **real-world ReAct Travel Assistant** that uses **live APIs** instead of hardcoded data.

Assignment Title

ReAct Travel Assistant – Real-Time Multi-Tool AI Agent

Objective

Build a **ReAct AI Agent** that can plan a trip using **real-time information**.

The agent must fetch:

- **Current date and time**
- **Real-time weather**
- **Live hotel data with pricing**
- **Transportation options (flight/train/bus)**
- **Travel summary with estimated budget**

The agent should follow the **Thought** → **Action** → **Observation** → **Final Answer** reasoning cycle.

Problem Statement

Create a conversational AI assistant that can answer queries such as:

I want to travel from Delhi to Goa from 20 Aug 2026 to 23 Aug 2026. Find the weather, available hotels with prices, transportation options, and estimate the total trip cost.

The assistant must **use tools** and **real APIs** instead of static responses.

Required Features

Feature	Requirement
Date & Time	Real current time
Weather	Live weather API
Hotels	Real hotel search API
Pricing	Return actual hotel prices
Transportation	Flight/train/bus information
ReAct Reasoning	Must show tool usage
Final Summary	Human-readable itinerary